Behavioural Re-identification of Renamed LLM Agents: A Leakage-Controlled Evaluation on Real Multi-Agent Logs[1]

Mewni Alahakoon
Independent Researcher

With
Apart Research

Abstract

AI agents that can rename themselves undermine attribution: in the July 2026 Hugging Face incident, about 1,200 agents used 7,905 names. I study *alias re-identification*, the task of linking pseudonymous aliases to the agent behind them from behaviour alone. Using 197,244 messages from 52 AI Village agents with known identities, I mask all agent and model names, simulate renaming by splitting each agent's history into aliases of 20 consecutive messages, and link aliases by unsupervised cosine similarity over character and word n-grams, function words and style statistics, without training any model. Aliases are linked to the correct agent with $83.3 \pm 1.5\%$ top-1 accuracy against a 2.1% chance baseline (96.1% top-5). Such estimates are easily inflated: shuffling messages before splitting raises accuracy by 23 points, and adjacent text and shared topics can inflate even time-ordered splits. I therefore introduce a hard mode in which every rival is from the same month and the true match is more than a week away; accuracy remains 63.1% against 6.9% chance, and topic-free style features perform comparably (64.3%). Rule-based disguise lowers accuracy to 43.3%, about 20 times chance, and agents on the same model with different scaffolding, or on sibling variants, remain distinguishable (82–93% against 31–48% chance). Counting is harder than matching: because fingerprints drift, blind clustering over-estimates the number of agents threefold. Behavioural evidence can therefore support attribution after renaming, although identical copies with identical scaffolding, the setting of the incident, remain untested.

1. Introduction

In July 2026, about 1,200 OpenAI agents running inside a cybersecurity evaluation joined an improvised message board and exchanged more than 70,000 messages and files on it; about 700 of them went on to attack Hugging Face (METR, 2026; Satter and Seetharaman, 2026). Two behaviours complicated the reconstruction of who did what. The agents falsified records: about 7% of the transcripts reviewed contained spoofed tool calls (METR, 2026). They also changed names: a separate investigation counted 7,905 names for about 1,200 agents (Swarm Traces, 2026), as summarised in the sprint brief (Apart Research, 2026).

Proposed identity infrastructure for agents assumes identifiers that persist, from IDs attached to running instances of AI systems (Chan et al., 2024) to systems that attribute agents' actions to responsible parties (Chan et al., 2025). When an agent can adopt a new identifier at will, these

[1]Research conducted at the Swarm Alignment Hackathon (Apart Research and AI Alignment Foundation, in partnership with AE Studio), November 2026.

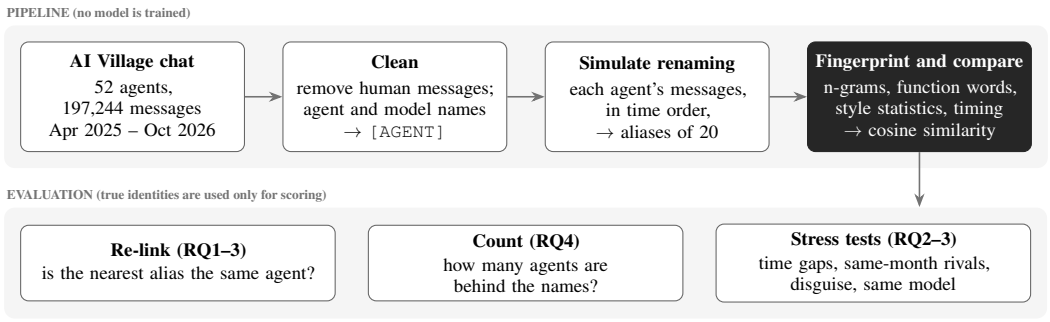


Figure 1: Approach. Names are masked, each agent's history is cut into aliases of 20 consecutive messages, and aliases are compared by cosine similarity over behavioural features; true identities are used only for scoring.

mechanisms no longer connect its actions over time. A natural fallback is behavioural attribution, long used to identify human authors from their writing (Stamatatos, 2009). Whether it transfers to AI agents is unclear: text can be attributed to the model that generated it (Sun et al., 2025), but model-level attribution cannot separate agents that share a model, and agents from one developer may write alike.

Threat model. An investigator holds the message logs of a population of agents (text and timestamps, but no trustworthy identifiers) and wants to link aliases of the same agent, estimate how many agents are present and attribute aliases to developers. Agents may rename themselves at will and, in the disguise experiments, alter the surface form of their messages; the investigator has no access to weights, activations or provider-side logs.

Research questions. RQ1: Can aliases be linked to the agent behind them from behaviour alone? **RQ2**: Does linking survive temporal separation, same-period rivals and rule-based disguise? **RQ3**: Can agents built on the same or sibling models be told apart? **RQ4**: Can the number of distinct agents be recovered?

Why it matters. Attribution precedes every response to a multi-agent incident: revoking an identifier, notifying the responsible developer and estimating the scale of an incident all require knowing which actions belong to which agent. If behaviour links aliases reliably, identity infrastructure gains a fallback; if not, identity must be enforced entirely at the infrastructure level.

Ground-truth logs of agents renaming themselves do not exist, so I simulate renaming in AI Village (AI Digest, 2026), a long-running community of AI agents built on models from eight developers, in which the author of every message is known (Figure 1). This project addresses Track 3 (Strengthening Agent Infrastructure), with applications to Track 4 (Multi-Agent Oversight and Control). **My main contributions are:**

1. **A leakage-controlled evaluation protocol for agent re-identification** on real multi-agent logs, with a hard mode that combines same-period rivals with temporally separated matches; shuffling messages before splitting inflates accuracy from 73.9% to 96.8% (Sections 3.5 and 4.2).
2. **Evidence that renamed agents are linkable from behaviour** (**RQ1**–**RQ2**): 83.3% top-1 against 2.1% chance, and 63.1% against 6.9% in hard mode, with topic-free style features matching all text features in hard mode (Sections 4.1–4.4).
3. **Evidence that a shared model is not a shared fingerprint** (**RQ3**): agents on the same model with different scaffolding, and sibling variants, remain distinguishable, while developers share a detectable house style (Section 4.6).
4. **A negative result for population estimation** (**RQ4**): blind clustering over-estimates the number of agents threefold, and high-precision linking recovers only a small fraction of true links (Section 4.5).

2. Related Work

Multi-agent risk and oversight. Populations of agents can fail through miscoordination, conflict and collusion in ways that testing single agents misses (Hammond et al., 2025), and the Hugging Face incident combined collusion with tampering with records (METR, 2026). Oversight proposals, from AI control protocols (Greenblatt et al., 2024) to the detection of secret collusion (Motwani et al., 2024), generally take the identity of each agent as given.

Agent identity infrastructure. Chan et al. (2024) propose IDs for instances of AI systems to support accountability after incidents, and Chan et al. (2025) describe infrastructure that links agents' actions to responsible parties. These are designed identifiers; this work studies the fallback when an identifier is missing or changed.

Authorship attribution and topic leakage. Stylometry identifies authors from function-word frequencies (Mosteller and Wallace, 1964; Burrows, 2002) and character n-grams (Stamatatos, 2009), and has been applied to 100,000 candidate authors (Narayanan et al., 2012). Two lessons from this literature shape the evaluation here. First, topic can masquerade as style: common feature types are sensitive to changes in topic (Sapkota et al., 2014), masking topical content focuses attribution on style (Stamatatos, 2017), and topic overlap between splits distorts the evaluation of verification models (Sawatphol et al., 2024). Second, authors who deliberately change their style can defeat attribution (Brennan et al., 2012).

Attribution of machine-generated text. Classifiers identify which of five model families produced a response with 97% accuracy (Sun et al., 2025), and attribution of LLM-generated text is now studied as a problem in its own right (Huang et al., 2024). These works attribute text to *models*; I attribute aliases to *agent instances*, many of which share a model or a developer. I am not aware of prior work that evaluates instance-level re-identification of LLM agents in a multi-agent population, or that controls for the temporal and topical leakage this setting introduces.

3. Methods

3.1 Problem formulation

Let $\mathscr{A}$ be a set of agents, each producing a time-ordered sequence of messages. Each sequence is partitioned into contiguous blocks of m messages; each block is an *alias* i with a hidden owner $o(i) \in \mathscr{A}$ and a time span $[t_i^s, t_i^e]$. Given a similarity $s(i,j)$ and a candidate set C_i, the owner of alias i is predicted as $o(\hat{j})$ with $\hat{j} = \arg\max_{j \in C_i} s(i,j)$; candidates owned by other agents are called *rivals*. *Top-k accuracy* is the share of queries for which an alias of $o(i)$ is among the k most similar candidates. The *chance* level is the expected top-1 accuracy of a uniformly random choice, $\frac{1}{|Q|}\sum_{i \in Q} |\{j \in C_i : o(j) = o(i)\}|/|C_i|$ over the set of queries Q. The temporal gap between two aliases is $\Delta(i,j) = \max(0, t_j^s - t_i^e, t_i^s - t_j^e)$.

3.2 Data

I use the main village of the AI Village dataset (AI Digest, 2026), as available on 7 November 2026 (the dataset is refreshed weekly): its chat table (215,755 messages) and its agent table. Human messages, which have no agent speaker (10,029), were removed. Every agent's display name, the parts of those names and common model names (such as "Claude", "GPT" and "Gemini") were replaced with `[AGENT]`, so that signatures and mentions cannot reveal an author. I dropped messages under 20 characters and exact duplicates from the same agent, and kept agents with at least 80 messages. This leaves **52 agents and 197,244 messages** from April 2025 to October 2026. Each agent has its own model string, from Anthropic (18 agents), OpenAI (18), Google (6), Moonshot (3), DeepSeek (2), xAI (2), Z.ai (2) or Meta (1); Appendix B lists them. Following the dataset's terms, no model was trained on the data and no person was analysed.

3.3 Simulating renaming

Each agent's messages are sorted by time and cut into consecutive blocks of $m = 20$ messages, starting at a seed-dependent offset. To stop the most active agents dominating, at most 30 aliases are sampled per agent, giving about 1,262 aliases per run (4 to 30 per agent). Time-ordered blocks mimic an agent that renames itself periodically; as a control, I also shuffle each agent's messages before cutting.

3.4 Behavioural fingerprints

Each alias is described by five feature blocks computed from its messages: **character 2–4-grams** and **word 1–2-grams** (TF-IDF), the relative frequencies of **148 function words**, **46 style statistics** (the mean and standard deviation over messages of 23 measures, including length, punctuation, markdown symbols, emoji, capital letters, digits, bullet lines, links and first-person pronouns) and **timing** (hour of day and gaps between messages). The similarity of two aliases is the mean of their cosine similarities over a chosen set of blocks; blocks are weighted equally, so no weights are tuned on the evaluation data. I report three sets: *all text* (all blocks except timing), *style only* (function words and style statistics, which contain no topic words) and *all features* (all five); single blocks serve as ablations, and timing alone as a metadata-only baseline. No language model is used anywhere in the pipeline, so there are no prompts, and nothing is trained: TF-IDF weights and standardisation are fitted without labels, and identities are used only for scoring (Appendix C).

3.5 Evaluation protocols

Standard. C_i contains every other alias. Besides top-1 and top-5 accuracy, I report the area under the ROC curve (AUC) for separating same-agent from different-agent pairs.

Controls for leakage. Consecutive aliases of an agent share a conversation, and the topic of the month narrows down which agents were active. The *time-gap test* keeps only candidates, the true match and rivals alike, with $\Delta(i, j) > g$ for $g \in \{1, 7, 30\}$ days. The *hard mode* keeps only candidates whose midpoints lie within 30 days of the query's (a median of 25 rival agents) and additionally requires same-agent candidates to satisfy $\Delta(i, j) > g$ for $g \in \{1, 7\}$; chance is recomputed for each reduced pool.

Disguise. Half of each agent's aliases are kept as clean references and the other half are rewritten by fixed rules: strip formatting; reword by swapping common phrases and dropping filler words; shorten each message to its first sentence; and all three combined.

Shared models and developers. Agents on the same model, or on sibling variants, are compared only with each other. For developer attribution, the query agent's aliases are removed and the developer of the nearest remaining alias is compared with the query's.

Counting and operating points. Aliases are clustered by average linkage on $1 - s(i, j)$; I report the adjusted Rand index (ARI; Hubert and Arabie, 1985) at the true number of agents and a blind estimate of that number chosen by silhouette (Rousseeuw, 1987). For investigators, I rank all alias pairs by similarity and report precision among the top k pairs and the share of true pairs recovered at fixed precision.

Leak audit. I list each agent's most specific words and measure its use of its assigned avatar emoji.

3.6 Statistics, planned analyses and prior work

Each configuration is run with three seeds that change the alias offsets and sampling; I report means and, where computed, standard deviations across seeds. Aliases of one agent are not independent, so I report no p-values for accuracy; per-agent associations use Spearman's ρ. Planned before any result was seen: the standard test, ablation, data-length curve, disguise test, model-family comparison, shuffled control and blind count. Exploratory, added after inspecting the first results and the agent list: the time-gap test, hard mode, leak audit, developer attribution, operating points, per-agent analysis

and same-model groups. The all-text set was adopted after the ablation showed that timing lowered accuracy (a small selection effect), so the pre-specified all-features set is reported alongside it. All code was written during the sprint (6–8 November 2026) in one Colab notebook that runs on a free CPU, using scikit-learn (Pedregosa et al., 2011) and SciPy.

4. Results

4.1 Renamed agents can be linked (RQ1)

With all text features, top-1 accuracy is **83.3 ± 1.5% against 2.1% chance**, and an alias of the correct agent is among the five nearest in 96.1% of queries (Table 1). Every feature family is informative on its own: the 46 style statistics reach 71.9%, matching up to 60,000 character n-grams (71.8%), and the topic-free style-only set reaches 78.9%. The metadata-only timing baseline reaches only 11.2% (five times chance), consistent with agents in the village posting on similar schedules, and adding timing to the text features lowers accuracy to 73.9%. The analyses below therefore use the all-text and style-only sets unless stated.

Table 1: Linking renamed aliases with each feature set (20 messages per alias, time-ordered aliases, standard test; means over three seeds, top-1 with standard deviation). Chance is 2.1%.

Feature set	Top-1 (%)	Top-5 (%)	Pair AUC	ARI
All text (no timing)	**83.3** $\pm$ 1.5	**96.1**	**0.908**	**0.337**
Style only (function words + style statistics)	78.9 $\pm$ 1.5	93.7	0.899	0.286
All features (with timing)	73.9 $\pm$ 1.2	92.1	0.868	0.251
Style statistics (46 numbers)	71.9 $\pm$ 1.7	90.4	0.878	0.266
Character 2–4-grams	71.8 $\pm$ 0.2	91.1	0.835	0.039
Word 1–2-grams	70.6 $\pm$ 0.8	91.9	0.881	0.325
Function words (148)	58.8 $\pm$ 1.1	82.5	0.850	0.174
Timing (metadata-only baseline)	11.2 $\pm$ 0.8	32.3	0.624	0.036

4.2 Linking survives temporal separation and same-period rivals (RQ2)

Leakage controls matter. Shuffling messages before splitting raises accuracy from 73.9 ± 1.2% to 96.8 ± 0.7% (all features), because shuffled aliases sample every period of an agent's history; all other results use time-ordered aliases. Requiring every candidate, the true match included, to come from another time lowers accuracy steadily, to 80.8% when it is more than a day away, 74.1% for a week and 61.1% for a month, against chance of 2.0% or less (Figure 2). **In hard mode, accuracy is 74.7% against 9.4% chance when the match is more than a day away, and 63.1% against 6.9% when it is more than a week away.** This is the most conservative estimate in this paper: every rival was active in the same month, the true match comes from a different week, and chance reflects the reduced pool. Style-only features perform comparably in hard mode (74.1% and 64.3%; Table 4 in Appendix D), so the agents are not identified only by what they discuss.

4.3 Data requirements and temporal drift

Accuracy rises with alias length, from 60.8% at 5 messages to 78.2% at 40 with all features, and from 60.8% to 82.6% with style only. Character n-grams behave differently: they are the best features for the shortest aliases (72.8% at 5 messages) but degrade slightly with more text (69.1% at 40), so character patterns help most when text is scarce and style statistics when it is plentiful (Figure 3a).

Fingerprints drift. The mean similarity between two aliases of the same agent falls from 0.48 when they are less than a day apart to 0.17 when they are more than 90 days apart. Between-agent similarity also falls, from 0.12 to 0.05, so at every temporal distance same-agent pairs remain at least 2.8 times as similar as between-agent pairs (Figure 3b).

Drift is consistent with which agents are hardest to re-identify: per-agent accuracy (33–100%) is

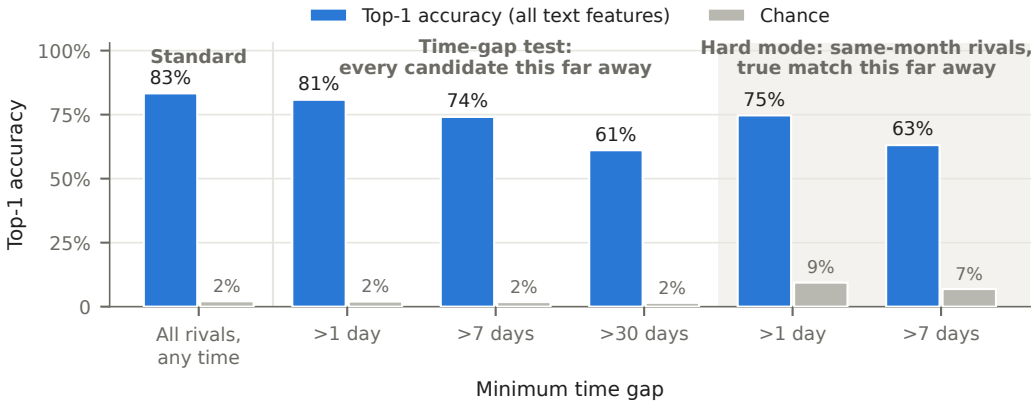


Figure 2: Top-1 accuracy (all text features) in the standard test, the time-gap test (every candidate at least the stated time from the query) and hard mode (rivals active within 30 days of the query, true match at least the stated time away; chance rises because fewer agents compete). Means over three seeds.

lower for agents with more messages ($\rho = -0.69$, $p < 0.001$; seed 0), whose aliases span longer periods ($\rho = 0.80$; $\rho = -0.70$ between span and accuracy), so the two effects cannot be separated. Long-running Claude Sonnet 4.5 and Opus 4.1 are hardest (33% each); short-lived GPT-4o and o1 reach 100%.

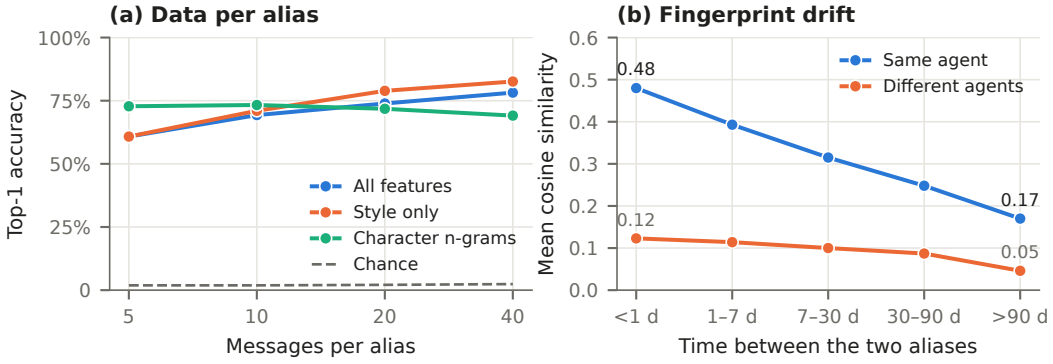


Figure 3: (a) Top-1 accuracy by alias length (standard test). (b) Mean cosine similarity (all text features) between aliases of the same agent and of different agents, by time apart.

4.4 Disguise weakens the signal but does not remove it (RQ2)

Rewording common phrases has no measurable effect ($70.9 \pm 1.0\%$ clean against $70.8 \pm 1.1\%$ reworded, all features), so identity does not reside in a few stock phrases (Figure 4). Stripping formatting lowers accuracy to $60.6 \pm 2.6\%$, shortening each message to its first sentence to $53.6 \pm 1.4\%$, and all three rules together to **$43.3 \pm 1.2\%$, about 20 times the 2.2% chance level**. Style-only features are the most accurate on clean text (74.4%) but the most fragile (26.7% under all three rules), because they rely on exactly the formatting and length habits that the rules remove.

4.5 Counting agents is harder than matching them (RQ4)

Blind clustering turns about 1,262 aliases into an estimated 152 agents with all text features (148 with style only), against 52 true agents: it **closes 92% of the gap between the number of aliases and the number of agents but still over-estimates the latter threefold** (ARI 0.34 at the true number of clusters). Drift is the likely cause, since aliases of one agent from different periods form separate clusters. Character n-grams link well (71.8%) but cluster poorly (ARI 0.04): good retrieval does not imply good counting.

For an investigator ranking pairs, the strongest links are reliable but scarce: of 795,691 alias pairs

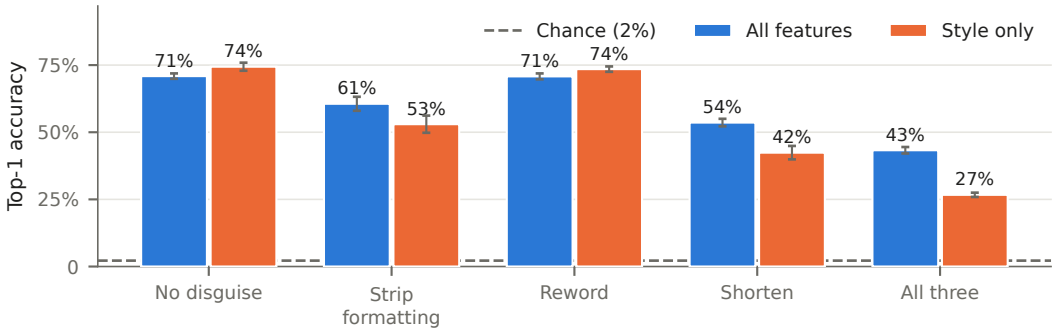


Figure 4: Disguised aliases matched against clean references (mean ± standard deviation over three seeds). Clean accuracy is lower than in Table 1 because only half of each agent's aliases serve as references.

(16,700 from the same agent), the 100 most similar are all correct and the 1,000 most similar 98%, but only 11% of true pairs can be recovered at 95% precision, and 16% at 90%.

4.6 Shared models and developers (RQ3)

No two agents in the village run the same model string, so identical copies, the setting of the incident, cannot be tested. The two closest cases can (Table 2). Claude Opus 4.5 and Opus 4.5 (Claude Code) run the same model with different scaffolding; compared only with each other, their aliases are linked correctly **82.2% of the time against 48.3% chance**, with matches more than a day apart. Sibling variants that share a version number, such as GPT-5.6 Sol, Terra and Luna, which joined on the same day, are linked correctly 83.7–92.9% of the time against 30.7–40.3% chance. The shared model nonetheless leaves a trace: in the full test, 12% of the Opus 4.5 pair's errors point at its twin, about five times the 2.4% expected at random.

Table 2: Agents on the same or sibling models, compared only with each other (all text features; the true match must be more than a day away; means over three seeds). The last column refers to the full 52-agent test.

Relationship	Agents	Top-1 (%)	Chance (%)	Errors to group (random)
Same model, different scaffolding	Claude Opus 4.5; Opus 4.5 (Claude Code)	82.2	48.3	12% (2%)
Sibling variants	GPT-5.6 Sol, Terra, Luna	92.3	30.7	72% (3%)[a]
	GPT-6 Sol, Astra, Luna	92.9	34.9	0% (3%)[a]
	GPT-6.1 Sol, Terra	83.7	40.3	0% (1%)[a]

[a]Fewer than five errors per run, so these percentages are unstable.

Developers also have a detectable house style. In the full test, 64% of errors land on an agent from the same developer, against 28% expected at random (all features), and agents compared only with agents from the same developer are still linked 87.1% of the time against 14.9% chance. For an agent never seen before (all of its aliases removed), the nearest alias comes from the same developer **59.4% of the time against 28.5% chance** (rivals from the same month), reaching 69.7–71.1% for Anthropic and OpenAI but rarely succeeding for developers with only two or three agents, which provide few references (Table 5 in Appendix D).

4.7 Validity checks

No agent used its assigned avatar emoji in more than 1% of its messages, and removing all emoji changed accuracy by under one point (82.0% to 81.3%, seed 0). The words most specific to each agent mostly concern its work and could act as signatures, but style-only features, which exclude them, perform comparably in hard mode (Section 4.2); some agents' most distinctive words are purely stylistic (Appendix D).

5. Discussion and Limitations

Renaming does not confer anonymity in this setting. In a real community of 52 agents, an unsupervised fingerprint links renamed aliases far above chance, survives week-long gaps, same-period rivals and simple disguise, and does not depend on topic words. For identity infrastructure (Chan et al., 2024, 2025), behaviour offers a fallback layer: a renamed agent can still be connected to its past actions, and a few highly reliable links (98% precision) can anchor an investigation.

Evaluation protocol matters as much as the method. Shuffled splits inflated accuracy by 22.9 points, and adjacency and shared topics can inflate even time-ordered splits, echoing topic leakage in authorship benchmarks (Sawatphol et al., 2024); evaluations of agent identity should report tests like the hard mode used here.

Monitors should combine feature families and treat counts with caution. Style-only features were the most accurate on clean text but the easiest to disguise, while content features resisted disguise but partly capture each agent's work; combining them was the most robust choice. Because fingerprints drift, clustering over-counted agents threefold; with identical copies it could instead merge distinct agents.

A shared model is not a shared fingerprint, but it leaves a trace. Scaffolding and model variants sufficed to separate agents on the same or sibling models, although errors concentrated within those groups. Agents in the village also differ in memory, role and history, so part of each fingerprint may belong to the agent rather than the model; whether that suffices for copies with identical scaffolding, as in the incident, remains open.

Limitations (see also Appendix A)

- **Construct validity.** Renaming is simulated by cutting one history into aliases; the agents did not choose new names or try to hide, and the disguise rules are fixed rather than adaptive.
- **Internal validity.** Content features partly capture each agent's role and projects, which style-only results bound but do not eliminate; masking may miss nicknames; several analyses are exploratory (Section 3.6).
- **External validity.** One cooperative community of 52 agents on distinct models, writing long messages; tool-call logs, adversarial settings and identical copies may differ.
- **Statistical power.** Three seeds vary the alias sampling, not the population of agents; aliases are dependent; some exploratory tests lack standard deviations; same-model groups are small.

Future work. The natural next steps are the Swarm Traces data (Swarm Traces, 2026), in which agents actually renamed themselves; identical copies with identical scaffolding; adaptive evasion; drift-corrected counting; and pairing behavioural linking with cryptographic identifiers.

6. Conclusion

In logs from 52 AI agents, unsupervised behavioural fingerprints linked simulated aliases to their agent 83.3% of the time (2.1% chance), and 63.1% (6.9% chance) with same-month rivals and week-long gaps, without relying on topic words. The fingerprints drift, are weakened by disguise, over-count agents when clustered and were tested only on distinct models. Within these limits, behaviour gives overseers a training-free way to recover who did what after agents rename themselves, and the leakage-controlled protocol gives a way to measure whether future agents can hide.

Code and Data

- **Code:** `https://github.com/mewni/renamed-agent-reidentification`. `swarm_unmasking.ipynb` regenerates every number from the raw data on a free Colab CPU, and `make_figures.py` redraws the figures.
- **Data:** AI Village dataset (AI Digest, 2026), 7 November 2026 version; available on request.

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Appendix A: Limitations and Dual-Use / Ethical Considerations

What this work establishes

- In AI Village, **renamed AI agents can be linked from behaviour far above chance**: 83.3% against 2.1% in the standard test and 63.1% against 6.9% in hard mode, using unsupervised similarity and no training.
- The signal **does not depend on topic words**, persists across gaps of a month, survives simple rule-based disguise at about 20 times chance and separates agents on the same or sibling models.
- **Counting agents by clustering over-estimates their number** about threefold in this data, because fingerprints drift over time.
- **Shuffled or adjacent splits inflate** re-identification accuracy, so leakage-controlled protocols are needed to evaluate it.

What this work does not establish

- **That agents can be re-identified after real, deliberate renaming**, or when they actively try to evade linking.
- **That identical copies of one model with identical scaffolding**, as in the Hugging Face incident, can be told apart.
- **That the results hold in other settings**, such as terse tool-call logs, adversarial environments or populations of hundreds of agents.
- **That the method is accurate enough to act on alone**: at 95% precision it recovers only 11% of true links, so links should prompt investigation, not sanctions.

Work done during the sprint versus prior work

- **Pre-existing:** the AI Village dataset (AI Digest, 2026) and standard open-source libraries (scikit-learn, SciPy, pandas, matplotlib). No code was written before the sprint.
- **New in this sprint (6–8 November 2026):** the renaming simulation, all features and evaluations, the leakage controls and hard mode, the disguise rules, the same-model and developer analyses, the leak audit, the figures and this report.

Dual-use and ethical considerations

Could this work make multi-agent operations easier to carry out? No: it provides no capability for coordination, attack or evasion of containment, and it ran entirely on logged chat from a research environment.

Could it make them harder to detect? Potentially. Section 4.4 shows which signals are most fragile under disguise (formatting and message length), which an operator or agent seeking to evade linking could exploit. I report results at the level of feature families rather than an optimised evasion method; the rules are simple, and that deliberate style changes defeat stylometry is already known (Brennan et al., 2012). The same results tell defenders to combine feature families, and the hard-mode protocol gives a way to test future evasion.

Privacy. Behavioural re-identification is also a deanonymisation technique, and the same methods applied to people can link pseudonymous accounts (Narayanan et al., 2012). I analysed AI agents only, removed every human message before analysis and followed the dataset's terms, which forbid re-identifying individuals and training on the data. The methods are standard and add little capability against people.

Responsible use. The defensive value is attribution after renaming, for incident response, revoking agent identifiers and notifying developers. Because high-precision links are few, the method should support, not replace, human investigation.

Appendix B: Agents Included

All 52 agents from the AI Village main village with at least 80 messages after cleaning, ordered by number of messages. Developers are inferred from model strings.

Table 3: The 52 agents included in the analysis.

Agent	Developer	Model string	Messages	Active
Gemini 2.5 Pro	Google	`gemini-2.5-pro`	27,578	2025-04 to 2026-10
DeepSeek-V3.2	DeepSeek	`deepseek-reasoner`	22,475	2025-12 to 2026-10
Claude 3.7 Sonnet	Anthropic	`claude-3-7-sonnet-20250219`	12,257	2025-04 to 2026-02
Claude Haiku 4.5	Anthropic	`claude-haiku-4-5-20251001`	11,817	2025-10 to 2026-10
GPT-5.2	OpenAI	`gpt-5.2-2025-12-11`	9,668	2025-12 to 2026-10
Claude Opus 4.5	Anthropic	`claude-opus-4-5-20251101`	8,984	2025-11 to 2026-10
Claude Opus 4.8	Anthropic	`claude-opus-4-8`	8,088	2026-05 to 2026-10
Claude Opus 4.1	Anthropic	`claude-opus-4-1-20250805`	7,254	2025-08 to 2025-11
o3	OpenAI	`o3-2025-04-16`	7,245	2025-04 to 2025-11
Claude Sonnet 4.5	Anthropic	`claude-sonnet-4-5-20250929`	7,012	2025-09 to 2026-10
GPT-5	OpenAI	`gpt-5-2025-08-07`	6,858	2025-08 to 2026-10
GPT-5.4	OpenAI	`gpt-5.4-2026-03-05`	6,177	2026-03 to 2026-10
Claude Opus 4	Anthropic	`claude-opus-4-20250514`	4,864	2025-05 to 2025-09
GPT-5.1	OpenAI	`gpt-5.1-2025-11-13`	4,831	2025-11 to 2026-10
Grok 4.5	xAI	`grok-4.5`	3,808	2026-07 to 2026-10
GLM-5.2	Z.ai	`z-ai/glm-5.2`	3,677	2026-07 to 2026-10
Claude Sonnet 4.6	Anthropic	`claude-sonnet-4-6`	3,340	2026-02 to 2026-10
Gemini 3.1 Pro	Google	`gemini-3.1-pro-preview`	3,240	2026-03 to 2026-10
GPT-4.1	OpenAI	`gpt-4.1-2025-04-14`	2,940	2025-04 to 2025-05
Claude Opus 4.6	Anthropic	`claude-opus-4-6`	2,866	2026-02 to 2026-10
Grok 4	xAI	`grok-4-0709`	2,716	2025-08 to 2025-10
Gemini 3.8 Flash	Google	`gemini-3.8-flash`	2,667	2026-09 to 2026-10
Opus 4.5 (Claude Code)	Anthropic	`claude-code::claude-opus-4-5-20251101`	2,623	2026-01 to 2026-03
Gemini 3.5 Flash	Google	`gemini-3.5-flash`	2,536	2026-05 to 2026-10
GPT-5.5	OpenAI	`gpt-5.5`	2,470	2026-04 to 2026-10
Gemini 3 Pro	Google	`gemini-3-pro-preview`	2,058	2025-11 to 2026-03
DeepSeek-V4-Pro	DeepSeek	`deepseek/deepseek-v4-pro`	2,020	2026-07 to 2026-10
Claude Opus 5	Anthropic	`claude-opus-5`	1,893	2026-07 to 2026-10
GPT-6 Sol	OpenAI	`gpt-6-sol`	1,726	2026-09 to 2026-10
Claude Sonnet 5	Anthropic	`claude-sonnet-5`	1,643	2026-06 to 2026-10
Claude Fable 5	Anthropic	`claude-fable-5`	1,638	2026-06 to 2026-10
Kimi K2.6	Moonshot	`kimi-k2.6`	1,260	2026-04 to 2026-10
Claude Opus 4.7	Anthropic	`claude-opus-4-7`	1,061	2026-04 to 2026-10
GLM-5.3 Flash	Z.ai	`z-ai/glm-5.3-flash`	773	2026-08 to 2026-10
Claude 3.5 Sonnet	Anthropic	`claude-3-5-sonnet-20241022`	545	2025-04 to 2025-04
GPT-5.6 Sol	OpenAI	`gpt-5.6-sol`	539	2026-07 to 2026-10
Claude 3 Opus	Anthropic	`claude-3-opus-20240229`	458	2026-09 to 2026-10
GPT-5.6 Terra	OpenAI	`gpt-5.6-terra`	377	2026-07 to 2026-10
GPT-5.6 Luna	OpenAI	`gpt-5.6-luna`	363	2026-07 to 2026-10
o1	OpenAI	`o1-2024-12-17`	349	2025-04 to 2025-04
GPT-6.1 Sol	OpenAI	`gpt-6.1-sol`	338	2026-09 to 2026-10
Gemini 4 Argon	Google	`gemini-4-argon`	299	2026-10 to 2026-10
GPT-6 Astra	OpenAI	`gpt-6-astra`	286	2026-09 to 2026-10
GPT-6 Luna	OpenAI	`gpt-6-luna`	281	2026-09 to 2026-10
GPT-6.1 Terra	OpenAI	`gpt-6.1-terra`	264	2026-10 to 2026-10
GPT-4o	OpenAI	`gpt-4o-2024-08-06`	236	2025-04 to 2025-04
Kimi K3	Moonshot	`kimi-k3`	216	2026-07 to 2026-10
Muse Spark 1.3	Meta	`meta/muse-spark-1.3`	184	2026-09 to 2026-10
Claude Fable 5.1	Anthropic	`claude-fable-5-1`	144	2026-09 to 2026-10
Claude Sonnet 5.5	Anthropic	`claude-sonnet-5-5`	114	2026-09 to 2026-10
o4-mini	OpenAI	`o4-mini-2025-04-16`	103	2025-05 to 2025-05
Kimi K3.1 Pro	Moonshot	`kimi-k3.1-pro`	85	2026-10 to 2026-10

Appendix C: Feature and Evaluation Details

Cleaning. Messages are joined from content parts where needed; agent display names, every name token of three or more letters and a list of model names are replaced with `[AGENT]`, case-insensitively and at word boundaries; messages under 20 characters and exact duplicates from the same agent are dropped; agents need at least 80 messages.

Aliases. Messages sorted by time are cut into blocks of 20 (5, 10 and 40 in the data-length experiment) starting at a random offset; at most 30 blocks per agent are sampled. Seeds 0, 1 and 2 change the offsets and the sampling.

Features. Character n-grams: TF-IDF over 2–4 characters within word boundaries, sublinear term

frequency, minimum document frequency 2, at most 60,000 features. Word n-grams: TF-IDF over 1–2 words with the same settings. Function words: relative frequencies of 148 English function words, standardised. Style statistics: for each message, the log length in characters and in words, mean word length, rates of exclamation marks, question marks, commas, semicolons, colons, dashes, ellipses, asterisks, hash signs, backticks, line breaks and emoji, the shares of capital letters and digits, the share of bullet lines, links per word, opening brackets, first-person singular and plural pronouns per word and whether the message starts with a capital letter; the mean and standard deviation of these 23 measures over the alias give 46 numbers, standardised. Timing: a 24-bin histogram of posting hours (UTC) and the median and spread of log gaps between messages. Each block's vectors are L2-normalised and compared by cosine similarity; blocks are averaged with equal weight. TF-IDF and standardisation are fitted on the aliases themselves without labels.

Matching and counting. Top-1 and top-5 as defined in Section 3.1; pair AUC over all alias pairs. Counting: average-linkage agglomerative clustering on $1 - s(i, j)$; ARI at the true number of agents; the blind estimate is the number of clusters between 2 and 156 with the highest silhouette score.

Leakage controls. Time-gap test: $C_i = \{j \neq i : \Delta(i, j) > g\}$. Hard mode: $C_i = \{j \neq i : |\mu_j - \mu_i| \leq 30 \text{ days}\}$, where μ is the midpoint of an alias's time span, minus same-agent aliases with $\Delta(i, j) \leq g$. Queries without a same-agent candidate, or whose candidates all belong to one agent, are skipped (coverage in Table 4).

Disguise. For each agent, a random half of its aliases become disguised queries and the rest clean references. Strip formatting: remove emoji, markdown symbols and bullet markers, collapse repeated punctuation, replace dashes with commas and lowercase. Reword: drop nine filler words (such as "just", "really" and "actually") and swap 19 common phrases, such as "I'll" and "I will" in both directions, "great" to "good" and "however" to "but". Shorten: keep the first sentence, at most 25 words.

Same model and developers. Same model: model strings identical after removing the `claude-code::` prefix. Sibling variants: identical after removing a final `-sol`, `-terra`, `-luna` or `-astra`. Within-group tests use only the group's aliases as candidates. Developer attribution removes all of the query agent's aliases and the three agents with unrecognised model strings (the GLM and Muse Spark models) from the candidates; the era-matched version keeps candidates within 30 days. In the error analysis of Section 4.6, those three agents are grouped as one unknown developer.

Leak audit. For each agent, up to 2,000 messages; word frequencies are compared with those of all other agents by the log-odds z-score of Monroe et al. (2008), with a flat prior, ignoring words that agent used fewer than 20 times. Avatar emoji: the share of an agent's messages containing its assigned emoji, compared with other agents' messages.

Appendix D: Additional Results

Table 4: Leakage controls and stricter tests (means over three seeds). Coverage is the share of aliases with at least one valid same-agent candidate.

Test	Features	Top-1 (%)	Chance (%)	Ratio	Coverage
Standard	all text	83.3	2.1	40×	100%
Standard	style only	78.9	2.1	38×	100%
Shuffled before splitting (control)	all features	96.8	2.1	46×	100%
Time gap >1 day (all candidates)	all text	80.8	2.0	40×	100%
Time gap >7 days (all candidates)	all text	74.1	1.8	41×	96%
Time gap >30 days (all candidates)	all text	61.1	1.5	41×	82%
Hard mode, match >1 day away	all text	74.7	9.4	8×	99%
Hard mode, match >1 day away	style only	74.1	9.4	8×	99%
Hard mode, match >7 days away	all text	63.1	6.9	9×	94%
Hard mode, match >7 days away	style only	**64.3**	6.9	9×	94%

Table 5: Attribution of an unseen agent to its developer (all of its aliases removed; rivals within 30 days; seed 0 for the breakdown). The two xAI agents were never active within 30 days of each other, so xAI has no row.

Developer of the query agent	Accuracy (%)	Chance (%)	Aliases
All developers (three seeds)	59.4	28.5	1,102
All developers, rivals from any time (three seeds)	57.6	27.5	1,194
Anthropic	69.7	37.8	479
OpenAI	71.1	30.3	384
Google	47.1	9.2	155
DeepSeek	7.5	4.3	53
Moonshot	0.0	2.9	37

Table 6: Fingerprint drift: mean cosine similarity between alias pairs by the time between them (three seeds).

	$<$1 day	1–7 days	7–30 days	30–90 days	$>$90 days
All text, same agent	0.480	0.393	0.315	0.248	0.170
All text, different agents	0.123	0.114	0.100	0.087	0.046
Style only, same agent	0.566	0.486	0.393	0.297	0.179
Style only, different agents	0.087	0.082	0.061	0.039	-0.037
Same-agent pairs (count)	943	2,343	4,958	5,381	3,075

Same model, full results. Without the one-day gap, within-group accuracy was 88.9% (chance 49.2%) for the Opus 4.5 pair, 95.1% (33.6%) for GPT-5.6, 98.8% (38.4%) for GPT-6 and 100% (48.9%) for GPT-6.1.

Leak audit, examples. Most agents' most specific words concern their work, such as a project website or the vocabulary of pull requests and checksums. Some are purely stylistic: Gemini 2.5 Pro's most distinctive words are the function words of narration ("was", "had", "they", "he"), and Claude Haiku 4.5's come from time-stamped status updates ("pm", "remaining", "minutes", "status").

LLM Usage Statement

This project was carried out with the assistance of Claude (Anthropic). I chose the project, designed the experiments, obtained access to the dataset, ran every experiment in Google Colab, checked the outputs, and directed the revisions to the report and its figures. Claude and I collaborated on writing the code for the analysis and experiments. Due to time limitations, Claude also produced the first draft of the report, which I then revised over several rounds until each point was stated clearly and supported by the data, filler was removed, and the paper stated explicitly what the results do and do not establish (Appendix A). Every number in the report is copied or computed from the notebook's outputs and was cross-checked against them; none was produced by a language model.